- Charles chart: how stable the original value is
- Independent indication of some sort of stability
- All we need to know is which things to start holding fixed more consistently each iteration
- If we held those things stable (probably in the core) that are fixed
- Slope of the dominant initial residue is somehow correlated with how steady it is under perturbation
- Way to measure
- Try the decoding order experiments again with additional residues
- Find most effective perturbation of

- You dont actually move the backbone at all
- Input into proteinmpnn that is more energetically based
- How do we know when a mutation is good? How do we know proteinMPNN is predicting a good system?

Charles: Expectation maximization algorith

- Hill climbing
- Proteinmpnn gives sequence backbone → alphafold or MD → new backbone → back into protein mpnn
- Run proteinmpnn on sequence → put back in molecular modeling → real life simulated annealing → higher affinity interaction

- Feeding all these relevant protein breathing structures for a protein mpnn and analyzing the sequences you get
- Laser mpnn?

https://www.science.org/doi/10.1126/science.add2187

- Might be original proteinmpnn paper

Goal

- Improve proteinMPNN
- Position at the top of math/cs

- Move away from trial and error
- Move toward statistical mechanics and information theory
    - **Statistical mechanics** = mathematical framework that applies statistical methods and probability theory to large assemblies of microscopic entities (mostly applied to physics)

    - **Information theory** = the mathematical study of quantifying, storing, and communicating information
        - **Shannon entropy** = measure of the expected value of the information content or uncertainty contained in a message
            - Measures the average uncertainty of a random variable X with probabilities $P(x_i)$
        - **Information rate / source doing** = efficiently representing information to remove redundancy (data compression)
        - **Channel capacity** = maximum rate at which information can be transmitted over a communication channel with low error, despite noise
        - **Mutual information** = measure of the amount of information that one random variable contains about another
- Gap in literature
    - No definitive benchmark paper that strictly analzes the math of a second pass or optimizes that general pass
- Existing literature
    - https://www.science.org/doi/10.1126/science.add2187
- Prove why second pass works better
- Relevant concepts
    - **Markov Chain Monte Carlo (MCMC) convergence**
        - MCMC methods = algorithms used to sample for complex high-dimensional probability distributions by constructing a markov chain that converges to the desires distribution over time
        - Monte carlo simulation = used to predict the probability of a variety of outcomes when the potential for random variables is present
            - Use repeated random sampling to obtain numerical results + approximate solutions for complex, uncertain, or deterministic problems
            - Simulate thoudsands of random inputs → provides probability distribution of potential outcomes
            - Modeling technique in finance that uses computational algorithms to simulate thousands of possible market scenarios to assess risk + uncertainty
            - Produces probability distribution of potential future outcomes
    - **Gibbs Sampling**
        - MCMC algorithm used to generate a sequence of samples from complex, high-dimensional, joint probability distributions
        - Instead of sampling directory from a joint distribution, it iteratively samples each variable from its conditional distribution, keeping all others fixed at their current values
        - Ultimately converges to the desired distribution
    - **Variational Inference**
        - ML technique that approximates complex, intractable posterior probability distributions by transforming inference into a simpler optimization problem

        - Aims to find a distribution q(z) within a chosen family that is closest to te true posterior p(z | x), usually by maximizing Evidence Lower Bound (ELBO) which minimizes Kullback-Leibler (KL) Convergence
            - ELBO = objective function used in variational inference to approximate intractable posterior distributions
    - Tractable posterior distn = bayesian posterior that can be computed, analyzed, and samples from directly, usually bc it has a known closed form mathematical expression
        - Occurs most often when using conjugate priors (posterior distn falls into same family as prior distribution
    - What does it mean when a probability distn has a closed form?
        - When its probability density function (PDF) or cumulative distribution function (CDF) can be expressed as a finite formula using standard mathematical functions
        - Avoids reliance on numerical approximations, infite series, or non-elementary insights
        - Implications
            - Analytical tractability = closed-form solutions allow for easier calculation of moments + parameter estimations

Iterative Gibbs Refinement of Autogressive Protein Potentials

- Fixed Point Iteration = numerical method used to find an approximate solution to an equation by rearranging it x = g(x)
    - Start with jinitial guess x_0, repeatedly apply formula $x_{k+1} = g(x_k)$
    - Process continues until sequence converges to a value p such that p = g(p) (the numbers aren't changing anymore)

Premise

- Proteinmpnn is an autoregressive model
- It approximates the conditional distribution $P(s_i \mid s_{<i}$, structure)
- The error in a single pass comes from the fact that s_i is decided without knowing $s_{>i}$ (this is an especially significant problem for earlier i residues)
- We propose a Refinement Pass where the model treats the first pass as a "noisy prior" and performs a series of updates to reach a global maximum of the joint probability P(S | structure)

The math

- **Information bottlenecks**
    - Analyze the mutual information between the input geometry and the output sequence
- **Error propagation analysis**
    - Use lipschitz continuity to prove how a small error in the first 5 residues of a sequence can lead to an exponential divergence in the final sequence recovery
    - A second pass contracts the error
- **Convergence proofs**

    - Show that under certain conditions (temperature T), the iterative pass is a contraction mapping (it is guaranteed to settle on a stable, optimal sequence)
        - And then we validate that the sequences are stable and optimal in the lab

The relevance: how you handle high-dimensional stochastic data
Need to emphasize

- Efficient sampling
    - Instead of just running the model twice, implement a **Metropolis-Hastings (MH)** step between passes
    - **MH algorithm** = an MCMC method that generates samples from a complex target distribution pi(x) by generating a candidate x' from a proposal q(x' | x) and accepting ith with probability alpha = min(fancy equation)
        - Allows sampling without knowing the normalization constant → cancels out in the acceptance ratio
    - Show you understand bayesian inference + computational efficiency
- Complexity Analysis
    - Provide strict Big-O bounds for the refinement algorithm
    - Show you can optimize for nanoseconds
- Ablation of Noise
    - Treat the first pass as a signal + noise problem
    - Use Signal processing terminology (e.g. signal-to-noise ratio in the logit space) to justify the second pass

I hate trial and error

- Focus on **Deterministic Systems** and **Statistical Guarantees**
- Trial and error is inefficient
- Seek to bound the problem mathematically before writing a single line of code

"Instead of running 1,000 simulations to see if a second pass helps, you define the problem so that the second pass *must* help." YES OMG YES PLEASE THIS IS ME

- **Fixed Point Theory** = studies conditions under which functions or mappings f(x) = x have at leat one invariant point (fixed point) in space
    - Combines topology, analysis, geometry to guarantee existence/uniqueness of solutions for differential equations, economics (nash equilibria), and numerical algorithms
    - Def: A point x* is a fixed point mapping f is applying the function yields the same point, i.e. f(x*) = x*
- **Contraction Mapping** = a function f mapping a metrix space (X, d) to itself, where all distances between points are reduced by a fxed factor k between 0 and 1.
    - Squeezes points close together

Mathematical Objective

- Treat proteinMPNN as a function f that maps a sequence space S to a probability distribution over that same space, conditioned on a fixed backbone B
    - So the sequence space is all the possible sequences that could occur
    - The size is just 21^(len of sequence)
- A single pass is just $s^{(1)} = f(B, \text{noise})$
- Second pass is the first step of an iterative process
    - $s^{(t+1)} = f(B, s^{(t)})$

Proof of Superiority

- Don't just show higher recovery
- Analyze the jacobian of the model
- Prove that **iterative refinement** is a contraction mapping on the metric space of protein sequences
    - Lipschitz constant of your refinement step is $L < 1$
    - By the Banach Fixed-Point Theorem, second pass is mathematically guaranteed to move closer to a unique, stable optimum (Fixed Point) than a single random sample
- Turns trial and error into a convergence proof

Algorithmic INformation Theory

- Can frame a second pass as a wa to reduce **relative entropy** (D_{KL})
- First pass has high uncertainty bc it lacks future context (purely autoregressive)
- Second pass uses first pass as a prior → increases the **Mutual Information** between the sequence and the fold
- Can quantity exactly how many bits of information the second pass adds to the system

From Gemini:

"I noticed that autoregressive models in high-dimensional spaces suffer from compounding error. To solve this, I re-engineered the inference step as a Gibbs-like iterative refinement. I proved that this transformed the sampling problem into a contraction mapping, which increased the information density of the output without needing a larger parameter set."

I LOVE ENGINEERING DETERMINISTIC SYSTEMS! I LIKE UNDERSTANIDNG EXACYLY WHY A SYSTEM WORKS! KNOWING THAT THE CODE WILL PRODUCE BETTER RESULTS BEFORE I RUN IT BECAUSE FIRST I PROVED THAT IT WILL MATHEMATICALLY! OMG YAY I LOVE THAT SO MUCH!

strip away the biology and treat the protein backbone as a fixed graph topology and the sequence as a stochastic state vector that we want to drive toward an equilibrium.